LAGUNA STATE POLYTECHNIC UNIVERSITY

Bachelor of Science in Information Technology

ITEP 414 – System Administration and Maintenance

Enterprise Infrastructure Planning for a Startup Company

Week 2 Portfolio Project

Prepared by: [Your Name]
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Date: [Submission Date]

Fictional client: ABC Startup Solutions

Project Overview

This report proposes an initial IT infrastructure for ABC Startup Solutions, a fictional software development startup with 20 employees on one office floor. Quantities and specifications are planning estimates and should be confirmed through a site survey, budget review, and vendor quotations before purchasing.

PART 1 — COMPANY PROFILE

Company Name

ABC Startup Solutions

Nature of Business

ABC Startup Solutions is a small software development company that builds websites, business applications, and digital solutions for local businesses. Employees need reliable computers, secure shared files, dependable internet, and a network that can grow with the company.

Company Vision

To become a trusted technology partner for small and medium-sized businesses by providing useful, reliable, and secure software solutions.

Company Mission

To develop practical software, protect customer and company information, support employees with dependable technology, and improve services through teamwork and continuous learning.

Office Location

Unit 3, Innovation Center, Mabini Avenue, Santa Cruz, Laguna, Philippines (fictional location for this project).

Organizational Structure

General Manager oversees four departments: Information Technology, Human Resources, Finance, and Sales. IT manages computers, servers, networks, user accounts, backups, and technical support. HR manages employee records, Finance handles accounting and payments, and Sales manages customer inquiries and sales activities.

Employee Distribution

Department	Employees	Main IT needs
Information Technology	5	Development workstations and admin tools
Human Resources	4	Employee records and restricted HR files
Finance	5	Accounting apps and secure records
Sales	6	Customer records and sales apps
TOTAL	20	One office floor

PART 2 — ENTERPRISE HARDWARE INVENTORY

Twenty desktops provide one primary workstation per employee. Five laptops support IT troubleshooting, meetings, and mobile work.

Asset ID	Hardware	Qty.	Department / Location	Purpose / Justification
HW-001	Business desktop computer	20	All departments	One primary workstation per employee; standardizes support.
HW-002	Business laptop	5	IT (3), management/shared (2)	Supports mobile work, troubleshooting, meetings, and backup workstations.
HW-003	Business server	1	IT / server room	Hosts selected internal services and shared files.
HW-004	Business router	1	Network closet	Routes traffic between the LAN and internet.
HW-005	Managed 24-port Gigabit switch	1	Network closet	Connects wired devices and supports VLANs.
HW-006	Network printer/scanner	2	HR/Finance and shared area	Supports confidential workflows and general printing.
HW-007	UPS, 1500 VA	2	Server/network closet	Provides short backup power and safe shutdown.
HW-008	Wi-Fi 6 access point	2	Office floor	Provides managed wireless coverage.
HW-009	NAS storage	1	IT / server room	Stores a second local copy of selected files and backups.
HW-010	Encrypted external backup drive	2	IT / secure storage	Rotating offline backups; one can be kept off-site.
HW-011	24-inch monitor	20	All departments	One monitor per desktop user.
HW-012	Keyboard and mouse set	20	All departments	Standard desktop input devices.
HW-013	24-port patch panel	1	Network closet	Organizes fixed network cabling.
HW-014	Lockable network cabinet	1	Network closet	Secures core network equipment.

PART 3 — ENTERPRISE SOFTWARE INVENTORY

Versions are stated as current stable or supported releases because versions change. Verify the latest release and license terms before deployment. AnyDesk business use requires an appropriate commercial license.

Software	Version	License	Purpose
Windows 11 Pro	Current supported release	Commercial license per device	Business desktop/laptop operating system.
Ubuntu Server LTS	Current supported LTS	Open-source; free to use	Stable platform for selected internal services.

Software	Version	License	Purpose
Microsoft 365 Apps / Office	Current subscription release	Commercial subscription	Documents, spreadsheets, presentations, and productivity.
Visual Studio Code	Current stable release	Free; open-source license	Code editor for software development.
Git	Current stable release	Open-source (GPLv2)	Tracks code changes and supports collaboration.
GitHub Desktop	Current stable release	Free app; GitHub terms apply	Graphical interface for common Git tasks.
Oracle VirtualBox	Current stable release	Base package GPLv3; extension pack separate terms	Runs test virtual machines.
Google Chrome	Current stable release	Freeware; Google terms apply	Web browsing and application testing.
Microsoft Defender Antivirus	Included with supported Windows	Included with Windows license	Built-in malware protection.
AnyDesk	Current version; business use only	Commercial license required for business use	Remote support when approved and controlled.
7-Zip	Current stable release	Open-source; see included license files	Compresses and extracts files.

PART 4 — ENTERPRISE NETWORK INVENTORY

Asset ID	Equipment / Service	Qty.	Location	Purpose / Justification
NW-001	Business fiber ISP modem/ONT	1	ISP handoff	Connects the office to the internet.
NW-002	Business router	1	Network closet	Routing, NAT, and VLAN gateway functions if supported.
NW-003	Business firewall	1	Internet edge	Filters traffic and separates network zones.
NW-004	Managed 24-port switch	1	Network closet	VLAN-aware wired connectivity.
NW-005	Wi-Fi 6 access point	2	Office floor	Separate corporate and guest wireless networks.
NW-006	24-port patch panel	1	Network closet	Terminates and labels fixed cable runs.
NW-007	CAT6 cable runs	30 estimated runs	Office floor	Connects workstations, printers, APs, and server.
NW-008	RJ45 connectors / keystone jacks	40 estimated	Cabling installation	Includes spare terminations for repairs.
NW-009	Cable labels and management	1 set	Network closet	Improves troubleshooting and maintenance.
NW-010	Business internet service	1 connection	Whole company	Target 300–500 Mbps fiber, subject to availability and budget.

PART 5 — ENTERPRISE NETWORK DIAGRAM

The ISP modem/ONT connects to the firewall, router/VLAN gateway, and managed switch. The switch connects the server/NAS, access points, and departmental devices. Departmental VLANs and guest Wi-Fi isolation reduce unnecessary access between groups.

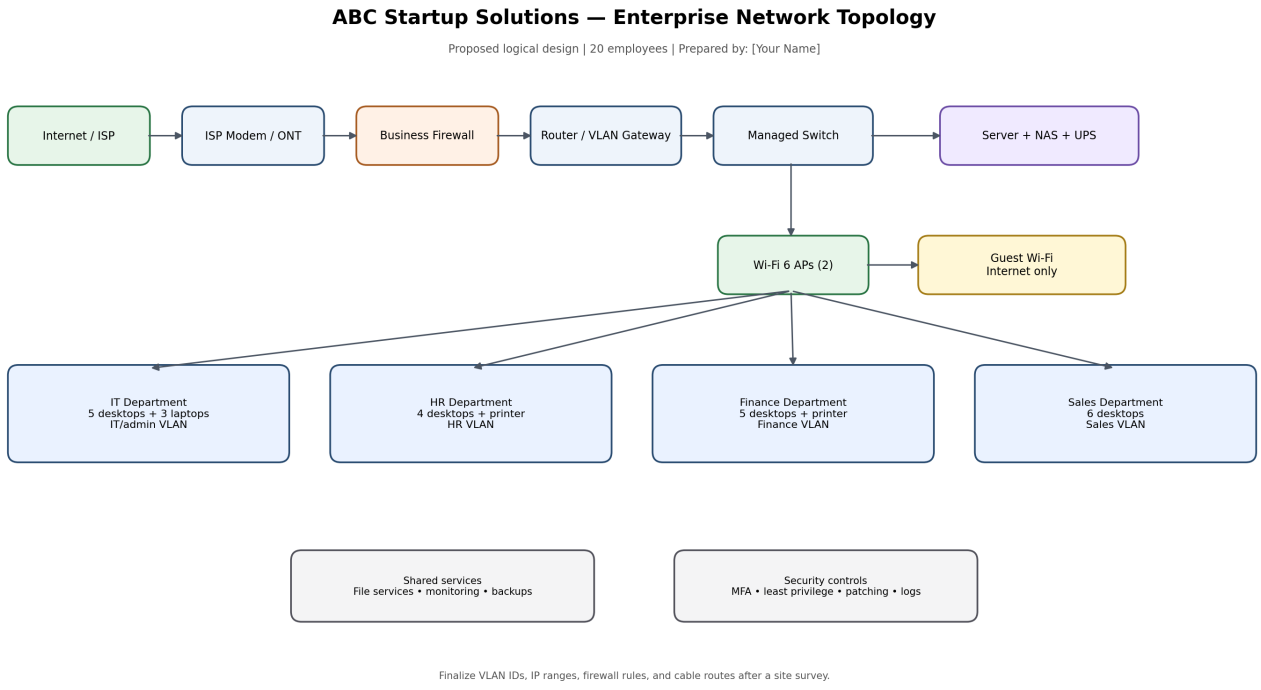


Figure 1. Proposed network topology for ABC Startup Solutions. Editable Draw.io source is included in the diagrams folder.

Suggested VLAN and IP Plan

Network	VLAN	Example subnet	Access policy
IT / Administration	10	192.168.10.0/24	Admin access for authorized IT only.
HR	20	192.168.20.0/24	HR resources for authorized HR users.
Finance	30	192.168.30.0/24	Finance resources restricted to approved users.
Sales	40	192.168.40.0/24	Approved business apps and shared services.
Servers	50	192.168.50.0/24	Required service ports from approved networks.

Network	VLAN	Example subnet	Access policy
Corporate Wi-Fi	60	192.168.60.0/24	Authenticated employee access.
Guest Wi-Fi	70	192.168.70.0/24	Internet only; block internal networks.
Network Management	99	192.168.99.0/24	Administrator-only device management.

These are example private IPv4 ranges, not a configured network. Confirm routing, DHCP, DNS, firewall rules, and possible VPN/cloud subnet overlap before implementation.

PART 6 — SYSTEM ADMINISTRATION ROLES

Helpdesk Technician

Responsibilities: receive support requests, troubleshoot hardware/software, set up accounts and devices, document incidents, and escalate complex issues. Skills: customer service, Windows basics, troubleshooting, clear documentation, and basic networking. Tools: ticketing system, remote-support tool, endpoint management console, and knowledge base. Possible certifications: CompTIA A+, Microsoft 365 Fundamentals, or Google IT Support Professional Certificate.

Network Administrator

Responsibilities: maintain switches, routers, Wi-Fi, IP addressing, VLANs, firewall rules, and network monitoring. Skills: TCP/IP, subnetting, DNS, DHCP, VLANs, wireless security, and troubleshooting. Tools: router/switch interfaces, Wireshark, ping/traceroute, SNMP monitoring, and firewall logs. Possible certifications: CompTIA Network+ or Cisco CCNA.

Linux System Administrator

Responsibilities: install and update Linux servers, manage users and permissions, monitor services and storage, automate tasks, secure configurations, and maintain backups. Skills: Linux command line, systemd, shell scripting, networking, logs, patching, and recovery. Tools: SSH, systemctl, journalctl, Bash, cron, and package managers. Possible certifications: Red Hat Certified System Administrator (RHCSA) or LPI LPIC-1.

Cloud Administrator

Responsibilities: manage cloud accounts, virtual machines, storage, identity/access, networking, monitoring, costs, and backups. Skills: cloud fundamentals, IAM, virtual networks, security, automation, and cost awareness. Tools: AWS Console/CLI, Azure portal/CLI, monitoring, and infrastructure-as-code tools. Possible certifications: AWS Certified Cloud Practitioner or Microsoft Certified: Azure Administrator Associate.

How the Four Professionals Work Together

These professionals work together to keep company technology reliable and secure. The helpdesk technician is usually the first contact and records problems. If an issue involves connectivity, the network administrator checks the router, switch, Wi-Fi, or firewall. If a server or Linux service fails, the Linux system administrator investigates the operating system, storage, permissions, and logs. For cloud services, the cloud administrator checks access, virtual machines, network rules, and costs. They share documentation, follow change procedures, communicate incidents, and escalate issues to the right person. This teamwork reduces downtime and protects company information.

PART 7 — INFRASTRUCTURE RECOMMENDATIONS

Internet Provider

Choose a local business fiber provider with reliable support, a clear repair process, and a service-level agreement if available. Aim for 300–500 Mbps symmetric fiber if affordable and available. Confirm coverage and actual prices before purchase. Consider a secondary connection if downtime would seriously affect operations.

Server Specifications

Start with a business-grade server with a 6–8 core CPU, 32–64 GB RAM (ECC if supported), mirrored SSDs for the operating system, and data storage sized to actual needs. Use Ubuntu Server LTS for selected internal services. Enable UPS-backed power, restricted administration, monitoring, and documented patching. Final sizing depends on which services will run on it.

Backup Strategy

Use a 3-2-1 approach: three copies of important data, on two types of storage, with one copy offline or off-site. Back up server data daily to the NAS, rotate encrypted external drives, and keep a protected off-site/cloud copy for critical files. Restrict deletion permissions and test restoration quarterly. RAID is not a backup by itself.

Security Recommendations

Use a business firewall, separate VLANs for departments, servers, management, and guest Wi-Fi, disable unused services, apply updates, use least-privilege accounts, encrypt laptops and backups, and enable MFA for email, cloud, and administrator accounts. Log important events and review access quarterly.

Antivirus / Endpoint Protection

Keep Microsoft Defender enabled and updated on Windows devices. Enable tamper protection and cloud-delivered protection where available. Review alerts and schedule scans. Avoid installing multiple real-time antivirus products that may conflict. Consider centrally managed endpoint protection as the company grows.

Password Policy

Require unique passwords or passphrases, use a password manager, and enable MFA wherever supported. Do not share accounts. Change passwords immediately if compromise is suspected. Disable accounts promptly when staff leave and review privileged access regularly.

Expansion Plan

Use a managed 24-port switch with spare ports, structured CAT6 cabling, and VLAN-capable equipment. Review capacity every six months. Add switch/AP capacity, server resources, or cloud services when employee count, traffic, storage, or availability needs increase. Document all changes and keep the asset inventory updated.

PART 8 — PERSONAL REFLECTION (DRAFT)

Personalize this reflection so it accurately describes your own experience before submitting.

This project helped me understand that system administration starts before computers and network devices are installed. A company needs a clear plan so that equipment, software, the network, and security controls support the way employees work. I learned that the number of employees and their departments affect the quantity of devices, the network design, and the access each group should have.

The most challenging task was deciding what equipment and network design would suit a small company without buying too much or leaving out important needs. The company needs enough computers for all 20 employees, but it also needs a server, switch, router, firewall, access points, printers, UPS units, and backup storage. These items have different purposes and must work together. Planning separate network areas for departments and guests was also important because not every device should have the same access to company information.

I also learned that security should be planned from the beginning. Strong passwords, multi-factor authentication, antivirus protection, regular updates, restricted permissions, and tested backups can reduce risks. Backups are especially important because files can be lost through hardware failure, accidental deletion, malware, or other incidents. Keeping a separate offline or off-site copy makes recovery more practical.

Planning before deployment helps an organization estimate costs, choose compatible equipment, avoid unnecessary purchases, and reduce installation problems. A network diagram and inventory also make it easier for another technician to understand the setup and troubleshoot issues. The plan should be reviewed as the company grows, because it may eventually need more storage, network ports, wireless coverage, or cloud services.

This project can help me become a better system administrator by improving my technical planning, documentation, research, and communication skills. It reminds me to explain why a recommendation is needed instead of simply listing devices. In a real workplace, I would confirm the budget, measure the office, check local internet availability, compare equipment prices, and discuss business needs with management before finalizing the design. I would also test the network, document changes, and regularly review security and backups. Overall, the project gave me a clearer picture of how different parts of IT infrastructure work together to support a business.

References

The assignment requirements are from the supplied ITEP 414 Week 2 PDF. The links below are starting points for checking software details, licenses, and example certifications.

• **Assignment brief:** ITEP 414 – System Administration and Maintenance, Week 2 Portfolio Project, supplied course PDF.

• **Microsoft Windows 11:** https://www.microsoft.com/windows/business

• **Ubuntu Server:** https://ubuntu.com/server

• **Microsoft 365:** https://www.microsoft.com/microsoft-365

• **Visual Studio Code:** https://code.visualstudio.com/

• **Git:** https://git-scm.com/

• **GitHub Desktop:** https://desktop.github.com/

• **Oracle VirtualBox:** https://www.virtualbox.org/

• **Google Chrome:** https://www.google.com/chrome/

• **Microsoft Security:** https://www.microsoft.com/en-us/security

• **AnyDesk licensing:** https://anydesk.com/en/pricing

• **7-Zip:** https://www.7-zip.org/

• **CompTIA certifications:** https://www.comptia.org/certifications

• **Cisco CCNA:** https://www.cisco.com/site/us/en/learn/training-certifications/certifications/enterprise/ccna/index.html

• **Red Hat RHCSA:** https://www.redhat.com/en/services/certification/rhcsa

• **Linux Professional Institute:** https://www.lpi.org/

• **AWS Certification:** https://aws.amazon.com/certification/

• **Microsoft Credentials:** https://learn.microsoft.com/credentials/

• **CISA Cybersecurity Performance Goals:** https://www.cisa.gov/cross-sector-cybersecurity-performance-goals

• **Draw.io / diagrams.net:** https://app.diagrams.net/

Submission Notes

• Replace all bracketed placeholders with your details.

• Review and personalize the reflection before submitting.

• Verify software versions, licenses, and vendor quotations before purchase.

- Check the network diagram against your instructor's expectations.
- Keep the editable DOCX and upload the README and diagram files to the required GitHub folders.